

A Rectifiable Case of Milman's Covering Property Conjecture

Automatic math-discovery packet

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Source Question

The source paper is Emanuel Milman, *Generalized Intersection Bodies*, arXiv:math/0512058. In Section 5 the paper formulates the following conjecture.

Covering Property Conjecture. For any $n > 0$, $1 \leq k \leq n - 1$, if $Z \subset G(n, n - k)$ is a closed set satisfying

$$\bigcup_{E \in Z} E \cap S^{n-1} = S^{n-1},$$

then there exists a measure $\mu \in \mathcal{M}_+(G(n, n - k))$, supported in Z , such that $R_{n-k}^*(d\mu) \geq 1$.

Claim

We prove the conjecture under the natural finite-rectifiability hypothesis suggested by Remark 5.12. If the closed covering set Z is countably k -rectifiable and has finite $\mathcal{H}^k$ -measure, then a scalar multiple of $\mathcal{H}^k|_Z$ gives the required finite positive measure.

Definitions and Normalizations

Put $m = n - k$ and $G = G(n, m)$. For $E \in G$, let σ_E denote the normalized rotation-invariant measure on $E \cap S^{n-1}$. If μ is a finite positive Borel measure on G , then

$$\langle R_m^* \mu, f \rangle = \int_G \int_{E \cap S^{n-1}} f(\theta) d\sigma_E(\theta) d\mu(E)$$

for every continuous f on S^{n-1} . The inequality $R_m^* \mu \geq 1$ means that $R_m^* \mu - \sigma$ is a positive measure on the sphere, up to the source paper's harmless normalization of the spherical measure σ .

Proof Intuition

The dimension count in the source's remark is exactly right. If Z has dimension k , then the incidence set

$$\Sigma_Z = \{(E, \theta) : E \in Z, \theta \in E \cap S^{n-1}\}$$

has dimension

$$k + (m - 1) = k + (n - k - 1) = n - 1,$$

the same as the sphere. The projection $\pi : \Sigma_Z \rightarrow S^{n-1}$, $\pi(E, \theta) = \theta$, is onto by the covering assumption. The area formula says that a Lipschitz map from an $(n - 1)$ -rectifiable set onto the

in the following sense:

Covering Property Conjecture. *For any $n > 0$, $1 \leq k \leq n - 1$, if $Z \subset G(n, n - k)$ is a closed set satisfying $\bigcup_{E \in Z} E \cap S^{n-1} = S^{n-1}$, then there exists a measure $\mu \in \mathcal{M}_+(G(n, n - k))$ supported in Z , such that $R_{n-k}^*(d\mu) \geq 1$.*

Under this conjecture, we immediately have the following counterpart to Lemma [5.10](#)

Lemma 5.11. *Assume the Covering Property Conjecture, and let $Z \subset G(n, n - k)$ be a closed set satisfying the covering property. Then there exists a measure $\mu \in \mathcal{M}_+(G(n, n - k))$ supported in Z , such that $R_{n-k}^*(d\mu) \geq 1$ and $R_k^*(d\mu^\perp) \geq 1$.*

Proof. Apply the Conjecture to the closed sets $Z \subset G(n, n - k)$ and $Z^\perp \subset G(n, k)$, and let $\mu_1 \in \mathcal{M}_+(G(n, n - k))$ and $\mu_2 \in \mathcal{M}_+(G(n, k))$ be the resulting measures. Then $\mu_1 + \mu_2^\perp$ is supported in Z and satisfies the requirements. $\square$

Remark 5.12. A very natural way to approach the proof of the Covering Property Conjecture, is to assume that the closed set Z satisfying $\bigcup_{E \in Z} E \cap S^{n-1} = S^{n-1}$ is minimal w.r.t. set inclusion (indeed, by Zorn's lemma it is easy to verify that there exists such a minimal set). The natural candidate for a measure supported on Z is simply the Hausdorff measure H_Z on Z , and it remains to show that H_Z is a finite measure and that $R_{n-k}^*(dH_Z) \geq \epsilon$ for some $\epsilon > 0$, using the minimality of Z . In particular, one has to show that the Hausdorff dimension of Z is k . Although having some progress in this direction, we have not been able to give a complete proof. We also remark that it is easy to construct

Figure 1: The Covering Property Conjecture and the beginning of Remark 5.12, page 37 of the source paper.

sphere must have positive Jacobian over almost every point of the sphere. Since this projection has uniformly bounded Jacobian, the pushforward of the natural incidence measure dominates a fixed positive multiple of spherical measure. But the pushforward is precisely $R_m^*(\mathcal{H}^k|_Z)$.

Rectifiable Covering Theorem

Theorem 1. *Let $1 \leq k \leq n-1$, put $m = n-k$, and let $Z \subset G(n, m)$ be compact. Assume:*

1. *Z is countably k -rectifiable;*
2. *$0 < \mathcal{H}^k(Z) < \infty$;*
3. *$\bigcup_{E \in Z} E \cap S^{n-1} = S^{n-1}$.*

Then there is a constant $c = c(n, k, Z) > 0$ such that

$$R_m^*(\mathcal{H}^k|_Z) \geq c\sigma.$$

Consequently, $\mu = c^{-1}\mathcal{H}^k|_Z$ is a finite positive measure supported in Z and satisfies $R_m^(\mu) \geq 1$.*

Proof. Let

$$\Sigma_Z = \{(E, \theta) \in Z \times S^{n-1} : \theta \in E\}.$$

Since Z is countably k -rectifiable and the unit-sphere bundle over the Grassmannian is smooth, Σ_Z is countably $(n-1)$ -rectifiable. The natural incidence measure

$$d\lambda(E, \theta) = d\mathcal{H}^k|_Z(E) d\sigma_E(\theta)$$

is finite and is locally comparable, with constants depending only on the standard Riemannian metrics and on n, k , to $\mathcal{H}^{n-1}$ on Σ_Z .

Let $\pi : \Sigma_Z \rightarrow S^{n-1}$ be $\pi(E, \theta) = \theta$. The covering assumption says that $\pi(\Sigma_Z) = S^{n-1}$. The map π is Lipschitz on the compact incidence bundle. Hence its approximate $(n-1)$ -Jacobian J_π is bounded above by some finite constant $L = L(n, k)$.

By the area formula for countably rectifiable sets,

$$\mathcal{H}^{n-1}(\pi(\Sigma_Z \cap \{J_\pi = 0\})) = 0.$$

Since $\pi(\Sigma_Z) = S^{n-1}$, it follows that for σ -almost every $\theta \in S^{n-1}$, there is at least one preimage $(E, \theta) \in \Sigma_Z$ with $J_\pi(E, \theta) > 0$. Applying the area formula again to the pushforward measure $\pi_\# \lambda$, its absolutely continuous part has density bounded below by a positive constant on almost every point of the sphere: each regular preimage contributes at least L^{-1} , up to the fixed comparability constants between λ and Hausdorff measure on Σ_Z . Thus there is $c > 0$ such that

$$\pi_\# \lambda \geq c\sigma.$$

Finally, for every non-negative continuous f on S^{n-1} ,

$$\begin{aligned} \int_{S^{n-1}} f d(\pi_\# \lambda) &= \int_{\Sigma_Z} f(\theta) d\lambda(E, \theta) \\ &= \int_Z \int_{E \cap S^{n-1}} f(\theta) d\sigma_E(\theta) d\mathcal{H}^k(E) \\ &= \langle R_m^*(\mathcal{H}^k|_Z), f \rangle. \end{aligned}$$

Therefore $R_m^*(\mathcal{H}^k|_Z) = \pi_\# \lambda \geq c\sigma$, as claimed. $\square$

Tangent Estimate Behind the Jacobian Bound

In a local chart at $E \in G(n, m)$, a tangent vector to the Grassmannian is a linear map $A : E \rightarrow E^\perp$. At an incidence point (E, θ) , a tangent vector to the fiber $E \cap S^{n-1}$ is $v \in E \cap \theta^\perp$, and the projection differential has the form

$$D\pi(A, v) = A\theta + v.$$

Since $A\theta \in E^\perp$ and $v \in E$, these terms are orthogonal, and

$$|D\pi(A, v)|^2 = |A\theta|^2 + |v|^2 \leq \|A\|_{\text{HS}}^2 + |v|^2.$$

This gives the uniform Lipschitz and Jacobian bound used above. The accompanying verifier script checks this dimension count and scalar estimate in random finite-dimensional samples.

Limitations and Novelty Check

This is not a proof of the full Covering Property Conjecture. It proves the conjecture for compact finite k -rectifiable covering sets. The remaining problem is to show that an arbitrary closed cover has, or can be reduced to, such a rectifiable finite-measure subcover. In particular, the theorem turns the route proposed in Remark 5.12 into a precise sufficient condition.

The same source paper also discusses the equivalence problem $\mathcal{BP}_k^n = \mathcal{I}_k^n$; this was later answered negatively by Milman’s arXiv:math/0701779 and is not the new result here. The lower dimensional Busemann-Petty endpoint cases and the linear inclusion questions $\mathcal{BP}_k^n \subset \mathcal{BP}_l^n$ and $\mathcal{I}_k^n \subset \mathcal{I}_l^n$ are not settled by this packet.

The run indexes were searched for arXiv:0512058, generalized intersection bodies, covering property, R_{n-k}^* , and the $\mathcal{BP}/\mathcal{I}$ questions. A bounded web search on July 18, 2026 for the exact phrase “Covering Property Conjecture” together with “Generalized Intersection Bodies”, “Radon”, and “Grassmannian” did not locate a later answer to this rectifiable case.

References

- E. Milman, *Generalized Intersection Bodies*, J. Funct. Anal. 240 (2006), 530–567; arXiv:math/0512058.
- E. Milman, *Generalized Intersection Bodies are not Equivalent*, arXiv:math/0701779.
- H. Federer, *Geometric Measure Theory*, Springer, 1969. Used for the area formula for rectifiable sets.
- P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge University Press, 1995. Standard reference for rectifiability and area/coarea formulae.

Human Review Recommendation

Review the measure normalizations in the definition of R_m^* , especially whether the source uses normalized or unnormalized Haar measure on $E \cap S^{n-1}$. This only changes the positive constant c , and the final measure can be rescaled. The main mathematical point to check is the rectifiable incidence-space application of the area formula.